

Supplementary Material for CauSample

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A Proof of the Causal Sampling Bound

We prove the individual-child recovery guarantee and conditional sampling-bound comparison in Theorem 4.2. We then establish a separate whole-batch guarantee for the offline procedure in Algorithm 1.

Proof roadmap. Lemma A.2 first identifies the rank of each local causal matrix and relates it to the DAG's roots and global rank. Proposition A.3 then bounds the number of random samples needed to preserve this rank. Given exact parent data, Lemma A.4 turns sampled full rank into unique coefficient recovery and exact child reconstruction. Combining these results proves the sampling bound; the submatrix rank relation gives the comparison under the stated additional condition. Finally, Corollary A.5 allocates failure probability across all children and uses induction along the causal orders to prove that Algorithm 1 returns the complete matrix with probability at least $1 - \delta$.

Notation and sampling model. Fix a batch $\mathbf{X}_i \in \mathbb{R}^{M \times T}$ and its SCS $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_K\}$. For a child metric m , let $\mathbf{P}_m = \mathbf{X}_i(\text{Pa}(m), :) \in \mathbb{R}^{\tau_m \times T}$, $\tau_m = |\text{Pa}(m)|$, and $\mathbf{C}_m = [\mathbf{P}_m^\top, \mathbf{x}_{m,i}^\top]$. We assume the exact within-batch relation $\mathbf{x}_{m,i} = \mathbf{w}_m \mathbf{P}_m$ and linearly independent parent rows. The coefficients may change between batches but are fixed within the batch under analysis. Let $\tau = \max_m \tau_m \geq 1$, $l_k = |\text{Root}_k|$, $l = \max_k l_k$, and $r = \text{rank}(\mathbf{X}_i)$.

For an orthonormal basis $\mathbf{U} \in \mathbb{R}^{T \times q}$ of a nonzero subspace, define its incoherence by $\mu(\mathbf{U}) = (T/q) \max_{1 \leq t \leq T} \|\mathbf{U}^\top \mathbf{e}_t\|_2^2$, where $\mathbf{e}_t$ is the t -th standard basis vector. The common upper bound μ applies to all parent row spaces and, when comparing with the global bound, to the row space of $\mathbf{X}_i$. Each metric samples d_m distinct time steps uniformly without replacement, independently of the fixed data. As in the main paper, $\Omega_m \in \{0, 1\}^{1 \times T}$ is the sampling vector. Write $\mathbf{D}_m = \text{diag}(\Omega_m)$, so $\mathbf{x}_{m,i,\Omega_m} = \mathbf{x}_{m,i} \mathbf{D}_m$ and the corresponding parent data are $\mathbf{P}_m \mathbf{D}_m$. All logarithms below are natural.

We restate Theorem 4.2 below.

THEOREM A.1 (CAUSAL SAMPLING BOUND). *For a batch $\mathbf{X}_i \in \mathbb{R}^{M \times T}$ and its SCS $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_K\}$, assume each child is an exact linear combination of its parents, whose vectors are linearly independent and span a subspace with incoherence at most μ . Let $\tau = \max_m |\text{Pa}(m)| \geq 1$, $l = \max_k |\text{Root}_k|$, and $\delta \in (0, 1)$. Then:*

- (1) *Given exact parent data, any child metric can be reconstructed exactly and its causal coefficients recovered with probability at least $1 - \delta$ by uniformly sampling d time steps without replacement, where $d \geq c\mu\tau \log(Kl/\delta)$ for a universal constant c ; if this requirement exceeds T , all entries are sampled.*
- (2) *If $Kl \leq r = \text{rank}(\mathbf{X}_i)$, the causal bound is no larger than the global-rank bound under the same incoherence bound: $c\mu\tau \log(Kl/\delta) \leq c\mu r \log(r/\delta)$.*

The first statement concerns an individual child given exact parents. Corollary A.5 gives the additional probability allocation required by the whole-batch argument below.

Offline causal matrix completion. The input to Algorithm 1 consists of the SCS, child sample budgets, and access to query entries of $\mathbf{X}_i$; unqueried entries are unavailable to the procedure. It processes each DAG in causal order, fully samples roots identified by $\text{Pa}(m) = \emptyset$, and reconstructs children from their samples and previously reconstructed parents. A rank check verifies whether the sampled parent data uniquely determine the coefficients. The output is a reconstructed matrix

or FAILURE if a rank check fails. Corollary A.5 specifies budgets that make the output equal to X_i with the stated probability.

LEMMA A.2 (LOCAL CAUSAL RANK AND ROOT SPAN). *For each child m in $\mathcal{G}_k$, let $r_k = \text{rank}(X_i(\pi_k, :))$. Then $\text{rank}(C_m) = \tau_m$ and*

$$\tau_m \leq r_k \leq l_k \leq l, \quad \tau \leq \max_k r_k \leq r. \quad (1)$$

PROOF. The parent matrix P_m has rank τ_m . Appending $\mathbf{x}_{m,i} = \mathbf{w}_m P_m$ adds no independent row, so $\text{rank}(C_m) = \tau_m$. Moreover, $C_m^\top$ is a row submatrix of $X_i(\pi_k, :)$, itself a row submatrix of X_i . Consequently, $\tau_m \leq r_k \leq r$.

Every row in a DAG lies in the span of its root rows: this is immediate for the roots, and follows for each child by substituting its parents' representations in topological order. Thus $r_k \leq l_k \leq l$. Taking maxima gives the remaining inequalities. The equality between local rank and parent count uses the stated linear independence assumption, not conditional independence alone. $\square$

The next proposition supplies the rank-preservation step used in subspace-based matrix completion [1].

PROPOSITION A.3 (RANK PRESERVATION BY RANDOM SAMPLING). *Let $\mathbf{B} \in \mathbb{R}^{p \times T}$ have rank $q \geq 1$ and a row space with incoherence at most μ . For a uniform sample of d distinct time steps, let $\mathbf{D} = \text{diag}(\Omega)$. If*

$$d \geq \min\{T, \lceil 8\mu q \log(q/\varepsilon) \rceil\}, \quad 0 < \varepsilon < 1, \quad (2)$$

then $\text{rank}(\mathbf{B}\mathbf{D}) = q$ with probability at least $1 - \varepsilon$.

Algorithm 1 Offline Causal Matrix Completion

Require: Query access to entries of $X_i \in \mathbb{R}^{M \times T}$; SCS $\mathcal{G}$ with orders π_k and parent sets $Pa(m)$; child sample budgets $1 \leq d_m \leq T$.

Ensure: Reconstructed matrix $\hat{X}_i$, or FAILURE.

1: Initialize $\hat{X}_i = \mathbf{0}$.

2: **for** each DAG $\mathcal{G}_k$ **do**

3: **for** each metric m in order π_k **do**

4: **if** $Pa(m) = \emptyset$ **then**

5: Collect all T values and set $\hat{X}_i(m, :) = X_i(m, :)$.

6: **else**

7: Draw d_m distinct time steps uniformly without replacement; encode them in Ω_m .

8: Set $\mathbf{D}_m = \text{diag}(\Omega_m)$; collect $\mathbf{y}_m = \mathbf{x}_{m,i} \mathbf{D}_m$.

9: Set $\hat{\mathbf{P}}_m = \hat{X}_i(Pa(m), :)$ and $\mathbf{A}_m = \hat{\mathbf{P}}_m \mathbf{D}_m$.

10: **if** $\text{rank}(\mathbf{A}_m) < |Pa(m)|$ **then**

11: **return** FAILURE.

12: **end if**

13: $\hat{\mathbf{w}}_m = \mathbf{y}_m \mathbf{A}_m^\dagger$.

14: $\hat{X}_i(m, :) = \hat{\mathbf{w}}_m \hat{\mathbf{P}}_m$.

15: **end if**

16: **end for**

17: **end for**

18: **return** $\hat{X}_i$.

PROOF. Full sampling gives the conclusion deterministically, so suppose $d < T$. Take a compact SVD $B = L\Sigma V^\top$, with $V \in \mathbb{R}^{T \times q}$ and $V^\top V = I_q$. Write $z_t = V^\top e_t$. Since $L\Sigma$ has full column rank, it suffices to show that the sampled Gram matrix

$$Y = V^\top D V = \sum_{t: \Omega(t)=1} z_t z_t^\top \quad (3)$$

is positive definite. Orthogonality and incoherence give $\sum_{t=1}^T z_t z_t^\top = I_q$, $\mathbb{E}[Y] = (d/T)I_q$, and $\lambda_{\max}(z_t z_t^\top) \leq \mu q/T$.

For independent uniform draws with replacement, the lower-tail matrix Chernoff inequality [3, Corollary 5.2 and Remark 5.3] bounds the probability that the smallest eigenvalue falls below its expectation. The trace-exponential comparison of Gross and Nesme [2] transfers this bound to uniform sampling without replacement: for any $\theta > 0$, the expectation of $\text{tr exp}(-\theta Y)$ is no larger than for draws with replacement. Hence, for $0 < \eta < 1$,

$$\Pr\{\lambda_{\min}(Y) \leq (1 - \eta)d/T\} \leq q \exp\left(-\frac{\eta^2 d}{2\mu q}\right). \quad (4)$$

With $\eta = 1/2$, Equation (2) yields

$$\Pr\{\lambda_{\min}(Y) \leq d/(2T)\} \leq q \exp\left(-\frac{d}{8\mu q}\right) \leq \varepsilon. \quad (5)$$

Outside this event, Y is positive definite, so $\text{rank}(BD) = q$. This argument uses a fixed sample count d , rather than the expected count of an independent Bernoulli sampling scheme. $\square$

LEMMA A.4 (RECOVERY OF CAUSAL COEFFICIENTS). *Given exact parent data, if $P_m D_m$ has full row rank, the sampled least-squares fit uniquely recovers w_m and reconstructs $x_{m,i}$ exactly.*

PROOF. Set $A_m = P_m D_m$ and $y_m = x_{m,i} D_m = w_m A_m$. The unique least-squares solution and its pseudoinverse expression are

$$\hat{w}_m = y_m A_m^\dagger, \quad A_m^\dagger = A_m^\top (A_m A_m^\top)^{-1}. \quad (6)$$

Full row rank makes the inverse exist and gives $A_m A_m^\dagger = I_{\tau_m}$. Therefore $\hat{w}_m = w_m$ and $\hat{x}_{m,i} = \hat{w}_m P_m = x_{m,i}$. These are the coefficient fit and reconstruction in Algorithm 1, lines 13–14, and the fit in Equation (7). Subspace membership alone does not ensure sampled full rank; Proposition A.3 supplies its probability guarantee. $\square$

PROOF OF THEOREM A.1. (1) *Sampling bound and reconstruction.* We prove the claim with $c = 8$; any larger universal constant also suffices. Consider first a child m in one DAG $\mathcal{G}_k$. Applying Proposition A.3 to its parent matrix, with $q = \tau_m$ and $\varepsilon = \delta$, gives the sufficient budget $8\mu\tau_m \log(\tau_m/\delta)$ before integer rounding and capping. Lemma A.4 then recovers the coefficients and the child trajectory. Because $\tau_m \leq l_k$ and $\tau_m \leq \tau$, the common budget $c\mu\tau \log(l_k/\delta)$ suffices for any fixed child in this DAG.

For K DAGs, assigning $\varepsilon = \delta/K$ to this rank test and using $l_k \leq l$ gives

$$8\mu\tau_m \log(K\tau_m/\delta) \leq c\mu\tau_m \log(Kl/\delta) \leq c\mu\tau \log(Kl/\delta). \quad (7)$$

Thus the theorem's budget $c\mu\tau \log(Kl/\delta)$ gives success probability at least $1 - \delta/K \geq 1 - \delta$ for any fixed child, given exact parents. Choosing the integer count $d = \min\{T, \lceil c\mu\tau \log(Kl/\delta) \rceil\}$ implements the sample requirement and the full-sampling case in the statement. Rounding upward preserves the guarantee, and full sampling recovers the child deterministically.

Equation (7) also shows that $d_m = \min\{T, \lceil c\mu\tau_m \log(Kl/\delta) \rceil\}$ suffices for a fixed child. This consequence supports the per-child frequency in Equation (8) under the randomized sampling

model. A union bound also covers one predetermined child from each DAG with joint probability at least $1 - \delta$; it does not, without further allocation, cover every child in every DAG.

(2) *Comparison with the global-rank bound.* The submatrix argument in Lemma A.2 establishes $\tau \leq r$. With the additional condition $Kl \leq r$,

$$c\mu\tau \log(Kl/\delta) \leq c\mu r \log(Kl/\delta) \leq c\mu r \log(r/\delta). \quad (8)$$

The comparison remains valid after integer rounding and capping at T . Local rank can be much smaller than global rank, but strict improvement is not asserted for every batch. This establishes both statements of Theorem A.1 and hence Theorem 4.2. $\square$

COROLLARY A.5 (CORRECTNESS OF ALGORITHM 1). *Under the stated exact linear relations, linear independence of parent vectors, and incoherence assumptions, let $N = M - \sum_k l_k \geq 1$ be the number of children and let $0 < \delta < 1$. If Algorithm 1 uses a budget $1 \leq d_m \leq T$ satisfying*

$$d_m \geq \min\{T, \lceil 8\mu\tau_m \log(N\tau_m/\delta) \rceil\} \quad (9)$$

for every child, then it returns $\hat{X}_i = X_i$ with probability at least $1 - \delta$.

PROOF. For analysis, assign every child its uniformly drawn sampling vector in advance, including children not reached if the algorithm stops. This is equivalent to drawing the vectors as they are needed in line 7, since the budgets are fixed and the draws do not depend on sampled values. For each child, define the event $\mathcal{E}_m = \{\text{rank}(\mathbf{P}_m \mathbf{D}_m) = \tau_m\}$ using its true parent matrix $\mathbf{P}_m$. Proposition A.3, with $q = \tau_m$ and $\varepsilon = \delta/N$, gives $\Pr(\mathcal{E}_m^c) \leq \delta/N$; if $d_m = T$, the rank is preserved deterministically. Thus the event $\mathcal{E} = \bigcap_{m \text{ child}} \mathcal{E}_m$ satisfies

$$\Pr(\mathcal{E}) \geq 1 - \sum_{m \text{ child}} \Pr(\mathcal{E}_m^c) \geq 1 - \delta. \quad (10)$$

The union bound requires no independence between these events.

On $\mathcal{E}$, we prove correctness by induction along each order π_k . Every root is collected in full and is therefore exact (line 5). Suppose all metrics preceding a child m have been reconstructed correctly. Since π_k is a causal order, all its parents precede it, so $\hat{\mathbf{P}}_m = \mathbf{P}_m$ in line 9. The event $\mathcal{E}_m$ makes $\mathbf{A}_m = \mathbf{P}_m \mathbf{D}_m$ full row rank, so the check in line 10 passes. Lemma A.4 then gives $\hat{\mathbf{w}}_m = \mathbf{w}_m$ and $\hat{X}_i(m, :) = X_i(m, :)$ in lines 13–14. Induction over all DAGs shows that the algorithm reaches line 18 and returns $\hat{X}_i = X_i$ on $\mathcal{E}$, which has probability at least $1 - \delta$. $\square$

Each completed run collects $T \sum_k l_k + \sum_{m \text{ child}} d_m$ entries; a run that reports failure may collect fewer. The child budget therefore does not describe the cost of every row. If $N = 0$, all metrics are roots and the algorithm recovers the matrix deterministically by full sampling.

Scope of the bound comparison. The exact DAG relations imply $r \leq \sum_k l_k \leq Kl$, so $Kl \leq r$ is a sufficient but restrictive condition: it requires independent root trajectories and $l_k = l$ for every DAG. It does not follow from sparsity alone. Without this condition, a comparison at a common failure tolerance ε still follows from $\tau_m \leq r$: $8\mu\tau_m \log(\tau_m/\varepsilon) \leq 8\mu r \log(r/\varepsilon)$. Both sides are sufficient budgets for known subspaces, not necessary sample counts or universal optimality claims. For $K = 1$, the theorem's bound is $c\mu\tau \log(l/\delta)$, and its comparison holds under $l \leq r$. For $K = M$, every DAG is a singleton and there are no child metrics; the causal bound is inapplicable, and the online framework uses temporal SMC for all metrics.

Extension to sampled roots. If roots instead use temporal SMC, their recovery events must also be included. Let $q_m = \text{rank}(\mathbf{H}_m)$ for an exactly known historical matrix, and assume the new root lies in its column span. With incoherence at most μ , assign roots the budget $\min\{T, \lceil 8\mu q_m \log(Mq_m/\delta) \rceil\}$ for $q_m \geq 1$, and children the corresponding budget with q_m replaced by τ_m . Proposition A.3, applied to $\mathbf{H}_m^\top$, preserves its sampled rank. Every sampled least-squares solution then gives the same root trajectory, since $\ker(\mathbf{D}_m \mathbf{H}_m) = \ker(\mathbf{H}_m)$. A union bound over at most M recovery events followed by topological induction gives whole-batch recovery with probability at least $1 - \delta$. A zero-rank historical space contains only the zero vector and needs no samples under this membership assumption. Counting only changes of the sampling vector cannot replace this argument: different parent spaces can fail on the same sample set.

Established guarantees and scope. Theorem A.1 establishes individual-child recovery and the conditional bound comparison; Corollary A.5 establishes exact matrix recovery by Algorithm 1 under the joint budgets in Equation (9). These guarantees concern uniform random sampling and the stated exact linear model. They do not automatically apply to the equal-interval schedule or to anomalies outside the parent subspace. Adding samples preserves an already full-rank sampled representation, but it does not by itself establish that rank or exact subspace membership. These conditions distinguish the sampling bound from the practical frequency and reconstruction rules.

B Anomaly-Model Learning and Updating

This appendix derives the approximate EM updates used by the Anomaly Sampler and Model Updater (Sections 4.3 and 4.6). The SCS fixes the allowed source metrics $m' \in \text{Pa}(m) \cup \{m\}$; only their propagation strengths and decay rates are learned. We use the exponential kernel $g_{mm'}(v) = a_{mm'} \beta_{mm'} e^{-\beta_{mm'} v}$ for $v > 0$ and the conditional event rate in Equation (9).

B.1 Likelihood and Event-Source Probabilities

Let h denote the final observed time step: $h = WT$ during offline learning and $h = iT$ after online batch i . The modeled history begins at time 0, and $E_m(h+1)$ includes all detected events of metric m through time h . For the continuous-time Hawkes model evaluated at these event times, the log-likelihood is

$$\ell(\theta) = \sum_{m=1}^M \left[\sum_{s \in E_m(h+1)} \log \lambda_m(s) - \int_0^h \lambda_m(t) dt \right]. \quad (11)$$

The event terms reward rates that explain the detected anomalies, while the integrated rate also accounts for time without events. For the exponential kernel, the integrated rate is

$$\int_0^h \lambda_m(t) dt = u_m h + \sum_{m' \in \text{Pa}(m) \cup \{m\}} a_{mm'} B_{mm'}(\beta_{mm'}; h), \quad (12)$$

where the finite-horizon contribution of source metric m' is

$$B_{mm'}(\beta; h) = \sum_{v \in E_{m'}(h+1)} \left(1 - e^{-\beta(h-v)} \right). \quad (13)$$

Following the branching interpretation of Hawkes processes [4], each target event has an unobserved source: spontaneous occurrence or triggering by an earlier self or parent event. With the parameters fixed at their current values, the E-step assigns the source probabilities

$$p_{ss}^m = \frac{u_m}{\lambda_m(s)}, \quad p_{sv}^{mm'} = \frac{g_{mm'}(s-v)}{\lambda_m(s)}, \quad v \in E_{m'}(s). \quad (14)$$

The allowed sources satisfy $m' \in Pa(m) \cup \{m\}$, and $E_{m'}(s)$ contains events strictly before s . These probabilities sum to one for each target event when $\lambda_m(s) > 0$.

B.2 Offline Parameter Updates

At each offline iteration, we recompute the source probabilities and the following statistics over the training events:

$$S_{u_m} = \sum_{s \in E_m(h+1)} p_{ss}^m, \quad S_{a_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}. \quad (15)$$

$$S_{\beta_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'} (s - v). \quad (16)$$

Here S_{u_m} is the expected number of spontaneous events, $S_{a_{mm'}}$ is the expected number triggered by metric m' , and $S_{\beta_{mm'}}$ is their probability-weighted total triggering delay.

Holding these probabilities fixed, the expected complete-data log-likelihood separates into a background term $Q_{u_m} = S_{u_m} \log u_m - u_m h$ and, for each allowed source–target pair,

$$Q_{mm'} = S_{a_{mm'}} (\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} B_{mm'}(\beta_{mm'}; h). \quad (17)$$

Terms independent of the updated parameters are omitted. The background update is $u_m = S_{u_m}/h$. For an identifiable source–target pair, the exact finite-horizon stationary conditions are

$$a_{mm'} = \frac{S_{a_{mm'}}}{B_{mm'}(\beta_{mm'}; h)}, \quad (18)$$

$$\frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} - a_{mm'} \sum_{v \in E_{m'}(h+1)} (h - v) e^{-\beta_{mm'}(h-v)} = 0. \quad (19)$$

Thus, exact finite-horizon updates couple the propagation strength and decay rate.

CauSample uses a boundary approximation to obtain explicit updates. Writing $n_{m'} = |E_{m'}(h+1)|$, we replace $B_{mm'}(\beta; h)$ by $n_{m'}$ and omit its derivative with respect to β . With the E-step statistics fixed, the approximate objective is $\tilde{Q}_{mm'} = S_{a_{mm'}} (\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} n_{m'}$. Setting its first derivatives, and that of Q_{u_m} , to zero gives

$$\frac{S_{u_m}}{u_m} - h = 0, \quad \frac{S_{a_{mm'}}}{a_{mm'}} - n_{m'} = 0, \quad \frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} = 0. \quad (20)$$

For positive, identifiable statistics these objectives have negative second derivatives, so the solutions maximize the approximate objective. The resulting approximate M-step is

$$u_m = \frac{S_{u_m}}{h}, \quad a_{mm'} = \frac{S_{a_{mm'}}}{n_{m'}}, \quad \beta_{mm'} = \frac{S_{a_{mm'}}}{S_{\beta_{mm'}}}. \quad (21)$$

These ratios apply when the denominators are positive and the event statistics identify the corresponding parameters. For a source–target pair with no positive triggering responsibility, $S_{a_{mm'}} = S_{\beta_{mm'}} = 0$; its decay rate is then unidentifiable, and the corresponding ratio does not apply. The approximation requires both the omitted kernel tails $\sum_v e^{-\beta_{mm'}(h-v)}$ and their derivative contributions to be small relative to the retained terms. A long observation horizon alone does not ensure this condition when many source events occur near its end. Consequently, these are approximate EM updates, rather than exact M-steps for the finite-horizon objective in Equation (17). We repeat the offline E- and M-steps with $h = WT$ until the maximum parameter change falls below ϵ .

B.3 Incremental Updates after Each Batch

We initialize the cumulative statistics from the final offline E-step. After batch i , each newly sampled anomaly at time s in metric m undergoes one E-step followed by one approximate M-step. Its source probabilities are computed using the current parameters and only events strictly before s , then added to the cumulative statistics:

$$S_{u_m} \leftarrow S_{u_m} + p_{ss}^m, \quad S_{a_{mm'}} \leftarrow S_{a_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}, \quad (22)$$

$$S_{\beta_{mm'}} \leftarrow S_{\beta_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'} (s - v). \quad (23)$$

For $m' \in Pa(m) \cup \{m\}$, we update the corresponding parameters using Equation (21) with $h = iT$ and $n_{m'} = |E_{m'}(iT + 1)|$. Both quantities include the initial training period and all subsequent batches. They remain fixed while processing the new events from the current batch; each parameter update precedes the E-step for the next event. The resulting parameters are used for predictions in the next batch.

Past source probabilities remain fixed, including contributions from batches that have left the sliding window. This cumulative update avoids repeated EM iterations over old target events, although each new target event still uses earlier parent and self events to compute its source probabilities. It does not claim exact maximization or monotonic improvement of the full-history likelihood. If a batch contains no sampled anomalies, no parameter update is performed; its elapsed time is included when the next event triggers an update. Within-batch predictions incorporate new events with fixed parameters, whereas parameter learning takes place after the batch ends.

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